Graph: ggtree and its main functions

Introduction

The `ggtree` package in R is a powerful tool for visualizing and annotating phylogenetic trees. It is built on top of the `ggplot2` package and allows for highly customizable and publication-ready tree visualizations.

Prerequisites

To follow this tutorial, ensure you have the following R packages installed:

```
install.packages("tidyverse")
if (!requireNamespace("BiocManager", quietly = TRUE)) {
  install.packages("BiocManager")
}
BiocManager::install("ggtree")
BiocManager::install("treeio")
```

We will use a `.nwk` dataset for demonstration.

```
library(ggtree)
library(treeio)
library(ggplot2)
library(dplyr)
```

Loading a Phylogenetic Tree

`ggtree` supports various formats of phylogenetic trees. The most common is the Newick format.

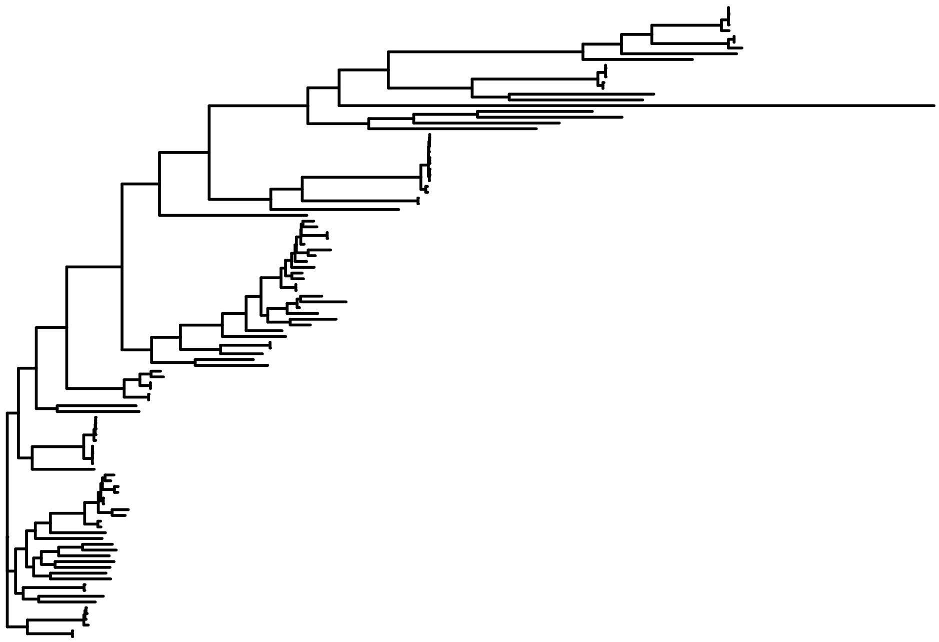
```r
# Load the tree from a Newick file
tree <- read.newick("example_2_phylo.nwk")

# Print tree structure
print(tree)
```

```
Phylogenetic tree with 110 tips and 108 internal nodes.

Tip labels:
  GCF_029224125.1, GCF_024584905.1, GCF_003176855.1, GCF_009867035.2, GCF_030285565.1, GCF_0
Node labels:
  , 0.981998, 0.987349, 0.946895, 0.994059, 1.000000, ...

Unrooted; includes branch lengths.
```

Basic Tree Visualization

Visualizing the tree is straightforward:

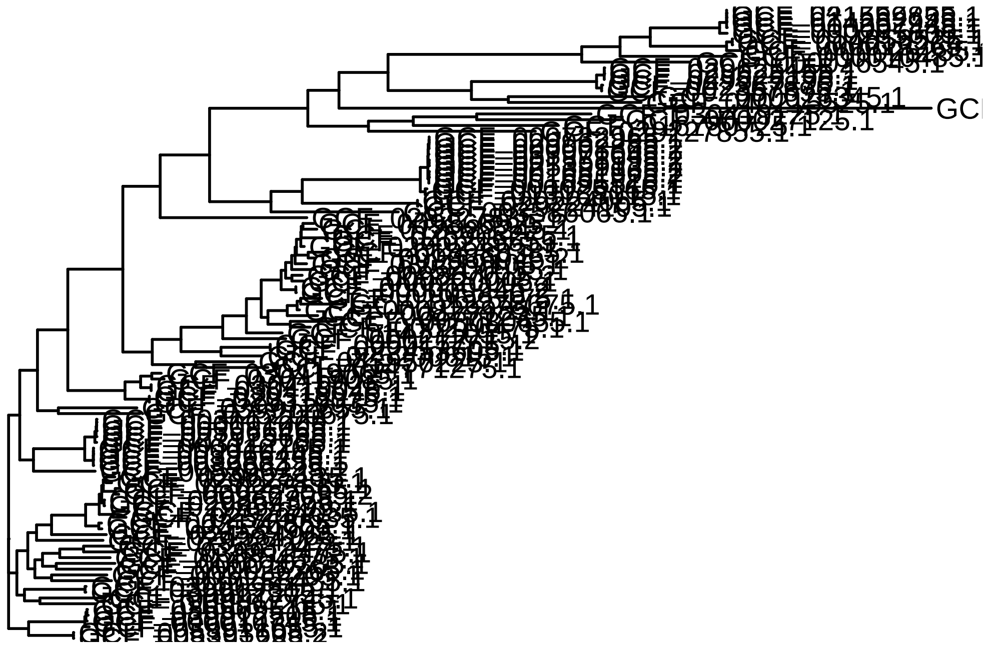
```r
# Basic tree plot
p <- ggtree(tree)
print(p)
```



Adding Tip Labels

You can add labels to the tips of the tree using `geom_tiplab`:

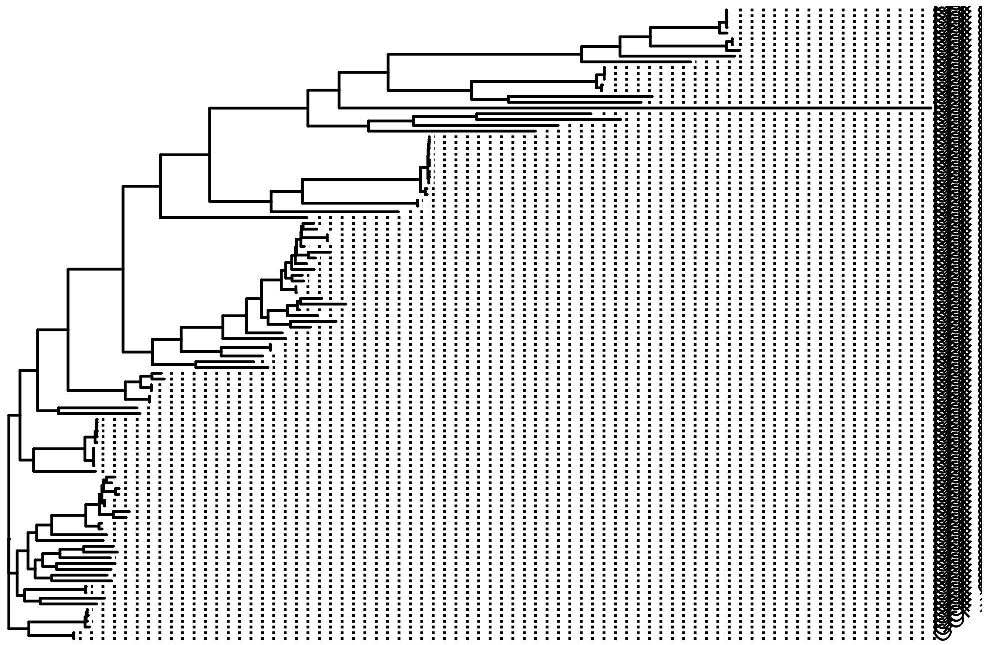
```
p <- ggtree(tree) +
  geom_tiplab()
print(p)
```



Parameters for `geom_tiplab`

- `size`: Adjusts the size of the labels.
- `align`: Aligns labels horizontally.
- `angle`: Rotates labels (useful for circular trees).

```
p <- ggtree(tree) +
  geom_tiplab(size = 3, align = TRUE, angle = 45)
print(p)
```

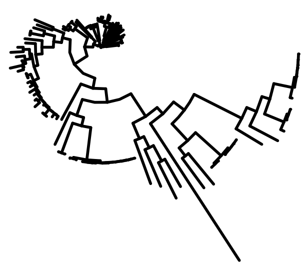


Tree Layouts

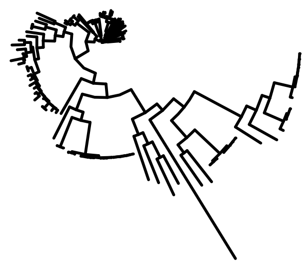
`ggtree` supports various layouts for trees, such as:

- `rectangular` (default)
- `circular`
- `fan`
- `slanted`
- `radial`

```
# Circular layout
p_circular <- ggtree(tree, layout = "circular")
print(p_circular)
```



```r
# Fan layout
p_fan <- ggtree(tree, layout = "fan")
print(p_fan)
```



Annotating Trees

Adding Node Labels

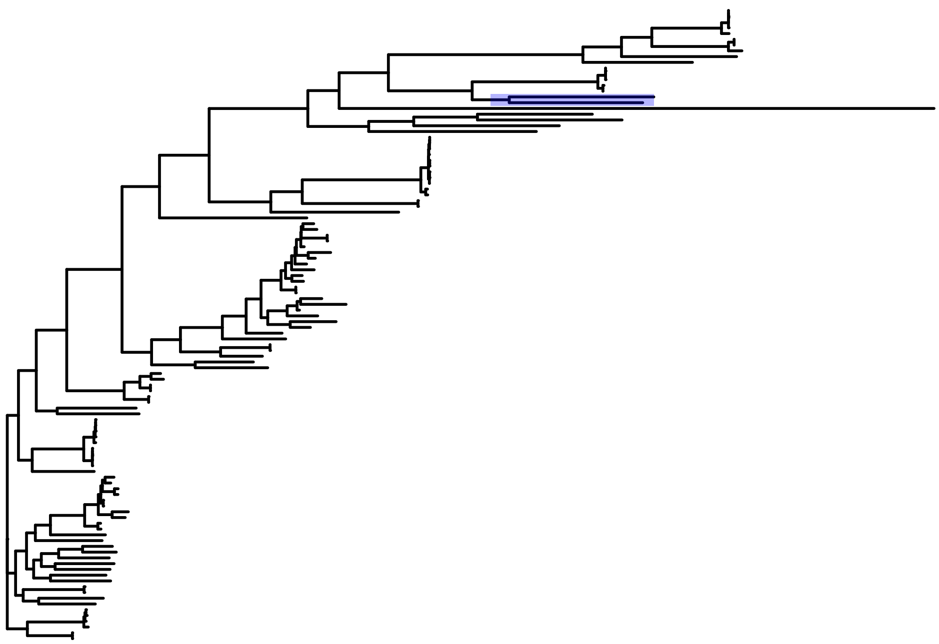
To annotate nodes, use `geom_nodelab`:

```r
p <- ggtree(tree) +
  geom_nodelab(aes(label = node), size = 3)
print(p)
```

Highlighting Clades

Highlight specific clades with `geom_hilight`:

```
p <- ggtree(tree) +
  geom_hilight(node = 199, fill = "blue", alpha = 0.3)
print(p)
```



Parameters for `geom_hilight`

- `node`: Specify the node ID to highlight.
- `fill`: The fill color for the highlight.
- `alpha`: Transparency of the highlight.

Annotating Specific Tips

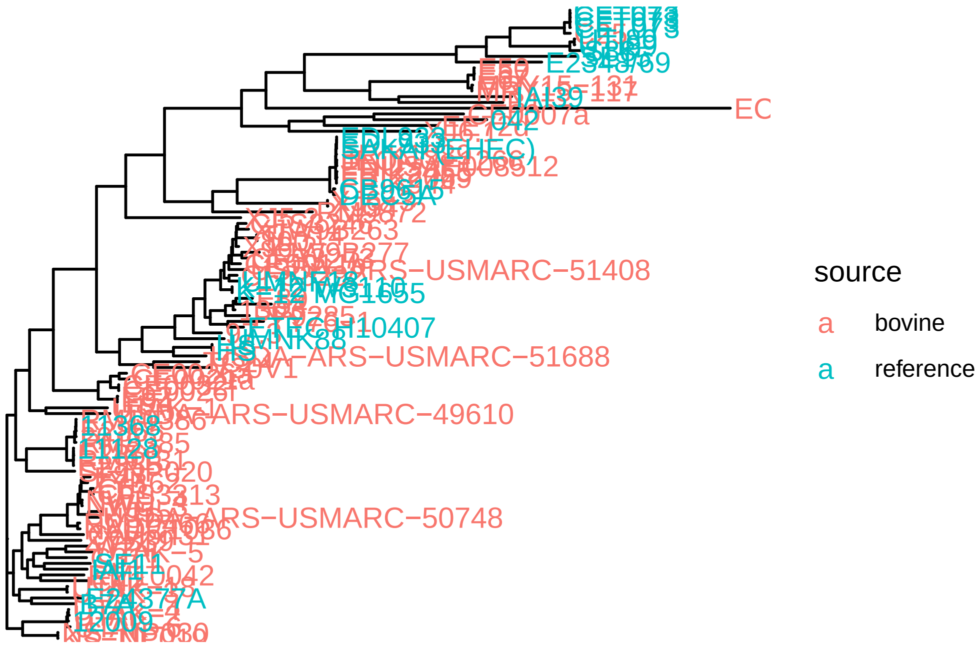
```
p <- ggtree(tree) +
  geom_tippoint(aes(color = label), size = 3)
print(p)
```

Adding Metadata

Metadata can be merged into the tree object and displayed, metadata is a `data.frame`:

```
# Cargar metadata
metadata <- readr::read_tsv("Experiment_Design_example_2.tsv")
metadata <- metadata %>%
  rename(label = Assembly.Accession)

# The rows have to be in the same order as the tree
metadata_ordered <- metadata %>%
  filter(label %in% tree$tip.label) %>%
  arrange(match(label, tree$tip.label))

metadata_ordered$Phylogroup <- replace(metadata_ordered$Phylogroup, is.na(metadata_ordered

metadata_ordered_2 <- metadata_ordered[,c(-1,-2)]
rownames(metadata_ordered_2) <- make.unique(metadata_ordered_2$Strain)
colnames(metadata_ordered_2)[1] <- "label"

# Change tip label
tree$tip.label <- metadata_ordered_2$label
```

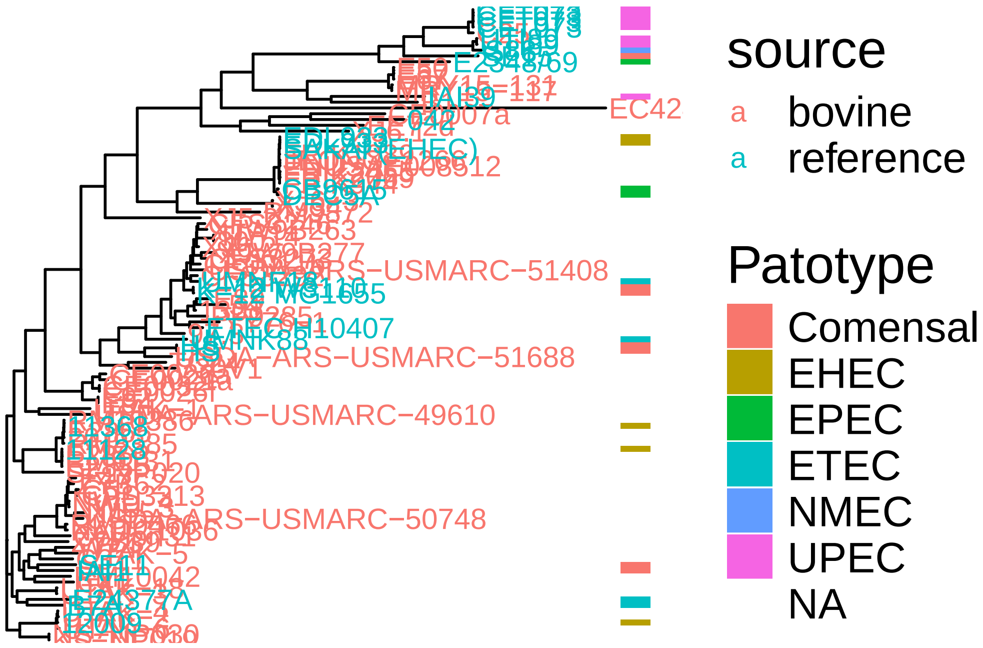
```
# Plot tree with metadata
p <- ggtree(tree) %<+% metadata_ordered_2 +
  geom_tiplab(aes(color = source))
print(p)
```



Creating Heatmaps

Use `gheatmap` to add heatmaps to the tree:

```
# Example matrix
set.seed(123)
matrix_data <- data.frame(Patotype = metadata_ordered_2$Patotype)
rownames(matrix_data) <- make.unique(metadata_ordered_2$label)

# Add heatmap
gheatmap(p, matrix_data, width=0.05,
         colnames=FALSE, legend_title="Patotype", color = NA, ) +
  theme(text = element_text(size = 20))
```


source
a bovine
a reference
Patotype
Comensal
EHEC
EPEC
ETEC
NMEC
UPEC
NA